

AIDAN MCLEAN

Keswick, ON

☎ (289) 716-7030 ✉ aidanmclean7030@yahoo.com [in linkedin.com/in/aidan-mclean-](https://www.linkedin.com/in/aidan-mclean-)

Education

McMaster University

September 2021 – April 2026

Bachelor of Applied Science, Honours Computer Science (Co-op)

Hamilton, ON

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Computational Theory
- Database Management
- Software Design
- Computer Networks
- Functional Programming
- Project Management

Work Experience

McMaster University Faculty of Engineering

May 2026 – Present

Research Assistant / Full-Stack Developer

Hamilton, ON

- Architected and developed a full-stack research platform (Node.js/Express, TypeScript, React, PostgreSQL) that allows researchers to submit and benchmark AI battery State-of-Charge models against one another, ranking submissions across 18+ evaluation metrics.
- Delivered features end to end across the stack, from REST API endpoints to the React components consuming them, including model submission workflows, results export, an in-app notification system, and an administrative management console.
- Designed the PostgreSQL schema (Drizzle ORM) covering authentication, model metadata, and evaluation history, and implemented cookie-based session authentication, rate limiting, and role-based access control across all protected routes.
- Built an asynchronous evaluation pipeline that runs each submission in an isolated Docker container, enabling concurrent scoring of multiple models without cross-submission interference, and delivers results through the notification system and email.
- Manage deployment on an Ubuntu server (Compute Canada's OpenStack cloud), including NGINX configuration, PM2 process management, and SSL/TLS certificate renewal; incorporate agentic AI development tools (e.g., Claude Code) into the workflow for debugging, code review, and infrastructure scripting.

Regional Municipality of York

May 2025 – September 2025

Software Developer Co-op Student

Newmarket, ON

- Customized Drupal Webforms with JavaScript to implement dynamic form behavior, validation, and improved user experience.
- Built front-end integrations with Microsoft APIM to apply business logic, route requests securely, and connect forms to backend services.
- Developed a TypeScript front-end application that integrated with an internal GIS map server to retrieve and visualize location-based data.
- Picked up Drupal, APIM, and Power Automate on the job and put them straight to use on live production forms.
- Documented APIs, workflows, and form logic so other developers could pick up the work without re-deriving it.

Regional Municipality of York

May 2022 – September 2022; May 2024 – September 2024

Customer Service System Support Analyst Student

Newmarket, ON

- Diagnosed and resolved technical issues for clients across a wide range of technical backgrounds, using targeted questions to narrow down root causes quickly.
- Logged detailed, readable tickets that held up for whoever picked up the issue next.
- Handled incoming calls, messages, and emails, prioritizing by urgency and impact.
- Installed and configured software for end users across the organization.

Regional Municipality of York

May 2023 – December 2023

Transit Fleet Analyst Co-op Student

Richmond Hill, ON

- Built Excel macros that automated recurring fleet reporting tasks, cutting down manual data entry.
- Wrote and maintained a PowerShell script that ran daily fueling-data analysis, catching issues in transit fleet operations before they became bigger problems.
- Supported YRT Fleet staff and end users on M5, Citrix, and Office issues.
- Tested M5 v23 ahead of release, checking for regressions across core fleet workflows.

Technical Proficiencies

Languages: Python, Java, TypeScript/JavaScript, SQL, C, Haskell

Frameworks/Tools: Node.js, Express, React, PostgreSQL, Docker, Git, NGINX, Linux